

BIOL3203 Food Microbiology

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Introduction

This course begins with a discussion of marine biology, covering the Introduction to Food Microbiology, Foodborne Pathogens, Detection and Enumeration of microbes in foods, Spore and their significant, Beneficial Bacteria, Factors that influence microbes in food and Physical Method of Food Preservation. This course aims to introduce the microscopic of food. Upon completion, students will be familiar with the mechanism and phenomenon in food microbiology.

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1 Lecture 1 Introduction to Food Microbiology & Food Pathogens

1. What is Food Microbiology ?

- **Microbiology** : Mikros = Small ; Bios = Life ; Logia = Study
- The study of small organisms
⇒ Mainly **unicellular** / **multicellular** organisms
- **Food Microbiology** : Interaction of microbes and foods
- The study of microbes in food + How food environments influence microbes

2. Three Broad areas

- (a) **Foodborne illness** and food microbial safety
⇒ Harmful to human
- (b) Microbe effects on **spoilage** and food quality
⇒ Harmful to food
- (c) **Beneficial use** of microorganism

3. History of Food Microbiology

- Robert Hooke : Publish microfungus
⇒ Also build early microscopes
- Antonie van Leeuwenhoek : First to observe and describe living bacteria
⇒ Use a higher magnification microscope
- Francesco Redi : First to disprove **spontaneous generation**
- Rudolf Virchow : Proposed the principle of **biogenesis**
- Louis Pasteur
 - Proved that Fermentation is Microbial
 - Disproved Spontaneous Generation
⇒ By having strong evidence for Biogenesis
⇒ Disproved by **Swan-neck Flask Experiment**
 - Developed Pasteurization
⇒ Heating can kill spoilage and pathogenic microbes

- Contributed to the development of **Germ Theory of Disease**
⇒ Means microbes can cause diseases

4. Spontaneous Generation

- Life arises spontaneously from **non-living matter**
- Example : Mice create from grains

5. Biogenesis

- Life comes from life
- All cells arise from pre-existing cells
- Disprove Spontaneous Generation : Maggot only appears on meat if flies land on it



Figure 1: Disprove Spontaneous Generation

6. Germ Theory of Disease : Specific microbes cause specific disease

7. **Swan-neck Flask Experiment**

(a) Boiling the broth

⇒ Kill all microorganisms

- Neck does not broken : Traps microbes
⇒ No contamination occurs
- Neck broken off : microbes reach the sterile broth and growth occurs
⇒ Contamination

(b) Conclusion 1 : **Microbes cause spoilage in air**

(c) Conclusion 2 : Develop canning and aseptic techniques

8. Roles of Microbes in food

(a) Beneficial Microbes

- Used in food production
⇒ E.g. Cheese, Alcohol
- Probiotics : Good Bacteria

(b) Spoilage Microbes

- Cause bad flavor, smell, appearance and texture
- Not harmful (attack) to human
- Attack foods

(c) Pathogens

- Causing foodborne disease
- Not visible
⇒ It may look, smell, taste normal
⇒ We don't know pathogen exist when we eat the food

(d) Bacteria

(e) Fungi : Molds & Yeast

(f) Virus

(g) Parasites

9. Naming System

- Binomial System : Genus + Species
- Written in *italics* or underlined
- Genus need to be capitalized
- Species remain lowercase
- We should write the full name when it is first used
- Strain Name : A genetic variant or subtype within same species
⇒ Only use when the microbe species have multiple strains

10. Bacteria

- Prokaryotic = Single Celled
- No true Nucleus
- Size : 1 – 3 μm
- Diverse shape
- Most of them are harmful (Foodborne Illnesses)
- Some of them are beneficial
- Classification
 - Shape
 - (a) **Cocci** = Spherical Shape
 - (b) Examples
 - i. *Staphylococcus aureus* : Clusters (staphylococci)
 - ii. *Streptococcus pyogenes* : Chains (streptococci)



Figure 2: Cocci Shaped

- (c) **Bacilli** = Cylindrical Shape = Rod Shaped
- (d) Examples
 - i. *Bacillus cereus*
 - ii. *Salmonella spp.*
 - iii. *Escherichia coli*
 - iv. *Listeria monocytogenes*

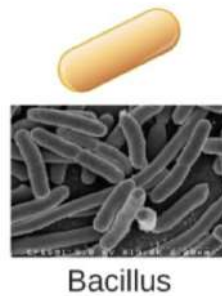


Figure 3: Bacillus Shaped

(e) **Spirilla** = Curved Shaped / Spiral Shaped

(f) Examples

- i. *Campylobacter jejuni* (Curved / Spiral Rods)
- ii. *Vibrio cholerae* (Comma Shaped)
- iii. *Vibrio parahaemolyticus* (Comma Shaped)

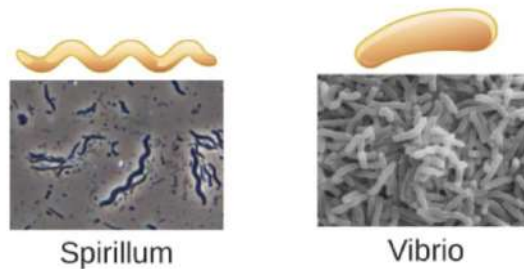


Figure 4: Spirilla Shaped

– Cell Wall Structure (Gram Stain)

(a) Gram Positive Bacteria

- * **Thick Peptidoglycan Layer** in cell wall
- * No Outer Membrane
- * More Resistant to physical stress
⇒ Drying & Salting
- * Gram Staining : Blue / Purple Colour
- * Example 1 : *Listeria monocytogenes*
- * Example 2 : *Staphylococcus aureus*



Figure 5: Gram Positive Bacteria

(b) Gram Negative Bacteria

- * **Thin Peptidoglycan Layer** in cell wall
- * Contain Outer Membrane
⇒ Outer Membrane containing **Lipopolysaccharide (LPS)**
- * Sensitive to drying
- * Gram Stain : Pink Colour
- * Example 1 : *Salmonella*
- * Example 2 : *E. coli*

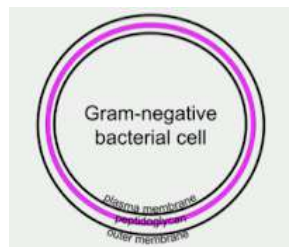


Figure 6: Gram Negative Bacteria

– Spore Formation

(a) Spore Formers

- * Produce **Endospore (Spores)** under **unfavorable conditions**
⇒ Highly resistant to harsh condition : Heat, Drying, Chemicals
⇒ **Dormant Structure**
- * Can survive after heating the food
⇒ Cause Food Spoilage and Foodborne Illness
- * Example 1 : *Clostridium*

* Example 2 : *Bacillus*

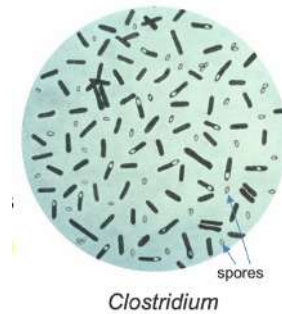


Figure 7: Spores

(b) Non-spore Formers

- * Do not produce **Endospore (Spores)**
⇒ This only exist at vegetative cells
- * Not good at against environmental stress
⇒ Can kill by **Pasteurization**
- * Example 1 : *Salmonella*
- * Example 2 : *Listeria*
- * Example 3 : *E. coli*

– Oxygen Requirement

(a) Aerobes

- * Require oxygen for growth
- * Example : *Pseudomonas spp.*
⇒ It can cause meat and fish **surface spoilage**
- * If put meat in anaerobic condition → ↑ *Clostridium spp.* Contamination Risk

(b) Anaerobes

- * Unable to grow when oxygen exists
- * Example : *Clostridium spp.*
⇒ Important in Canned & Vacuum Packed Products

(c) Facultative Anaerobes

- * Can grow with or without oxygen
⇒ But more prefer and like oxygen

- * Example : Most of the pathogens
- (d) Microaerophiles
 - * Require low oxygen
 - * Grow just below the surface medium but not at the top part
 - * Example : *Camplobacter spp.*
- Temperature Preference
 - (a) Psychrophiles
 - * Love cold temperature but more sensitive in warm temperature ($> 20^{\circ}\text{C}$)
 - * Seldom cause food illness and spoilage
 - (b) Psychrotrophs
 - * Cold tolerant
 - $\Rightarrow$ Grows in the fridge
 - * Grow best at warmer temperature
 - * Spoilage of refrigerated food *Pseudomonas spp.* and pathogen in refrigerated food (*Listeria*)
 - (c) Mesophiles
 - * Grows at moderate temperature
 - * Most of the pathogens
 - (d) Thermophiles
 - * Grow rapidly at relatively high temperature ($> 40^{\circ}\text{C}$)
 - * Often important in spoilage of thermally processed food
 - $\Rightarrow$ E.g. Canned Foods

11. Fungi

- (a) Yeasts
 - Unicellular Eukaryotes
 - $\Rightarrow$ Size : $>$ Bacteria
 - Most are **facultative anaerobes**
 - Cause food spoilage
 - $\Rightarrow$ E.g. Fruit Juice, Wine, Jam, Cheese
 - $\Rightarrow$ Sugar Fermentation $\rightarrow$ Gas / Alcohol / Off-flavors

- Advantage : Fermentation in food production

(b) Molds

- Multicellular Eukaryotes
⇒ Size : > Yeasts
- Produce **Spores**
⇒ Used for **Reproduction**
- Cause food spoilage
⇒ Grow in very harsh condition (Even bacteria cannot survive)
⇒ Fuzzy Patches & Colored Spores, Soft Texture, Off-flavors
∴ Degradation
- Some molds can produce **Mycotoxin**
⇒ Harmful to human



Branching thread-like structure (**hyphae**)

Figure 8: Hyphae

12. Viruses

- Acellular = Not a living cell
- Can only replicate inside living host cells
- No growth in food
⇒ Can survive in food (Lying on top of food)
⇒ Food is the cargo to let virus enter living bodies
- Need an animal host to grow
- We should prevent contamination
⇒ Not prevent the growth
- Cause foodborne illness

- Example 1 : Norovirus
- Example 2 : Hepatitis A

13. Parasites

- Protozoa : Unicellular Eukaryotic
- Worms : Multicellular Eukaryotic
- Need a living animal host to reproduce
- We should prevent contamination
⇒ Not prevent the growth
- No growth in food
⇒ Can survive in food in the form of **cysts** or **larvae**
⇒ Both have different mechanism
- Example 1 : *Giardia* (Stay at water)
- Example 2 : *Trichinella* (Stay at uncooked pork)

Roles of Microorganisms in Food

Role	Bacteria	Yeasts	Molds	Viruses	Parasites
Foodborne Illness	✓	✗	✓*	✓	✓
Food Spoilage	✓	✓	✓	? ✗	? ✗
Food fermentation	✓	✓	✓	? ✗	? ✗

* Note: Mold itself is rarely an infectious pathogen, but some can produce mycotoxins which cause illness.

Figure 9: Roles of Microorganisms in Food

14. Behavior of Microorganisms in Food

- Depending on the storage condition & properties of food
⇒ Microbes can grow, survive, die
⇒ Viruses and Parasites do not grow in food
- Growth (Multiplication)
 - Increase in cell number by **Binary Fission**
 - Only occurs under favorable condition
- Survival
 - Cell Number Unchanged = No Replication

- No Growth and No Death
- Bacteria still alive (Viable)
⇒ Can back to growth once condition is favorable

(d) Death (Inactivation)

- Decrease in cell number
- Bacteria loses its ability to multiply **irreversibly**

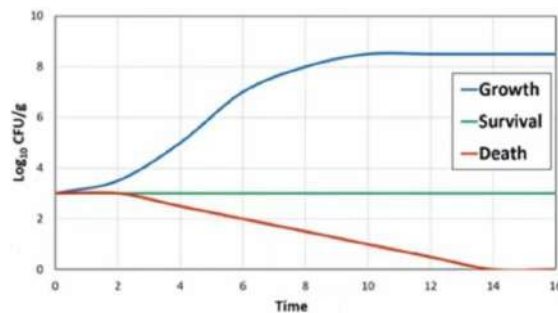


Figure 10: Behavior of microbes

15. Four Phase of Microbial Growth in Food

- (a) Lag Phase : Cells adapt to new environment
- (b) Logarithmic / Exponential Growth Phase : Bacteria reproduce rapidly by binary fission
⇒ Food become unsafe or spoiled (Bacteria is out of control)
- (c) Stationary Phase : Growth decreases + Nutrients become limited and waste products accumulate
- (d) Death Phase : Cell Death Rate > Growth Rate
- (e) Slope = Growth Rate
- (f) Food Preservation : Keep bacteria in Lag phase without growth

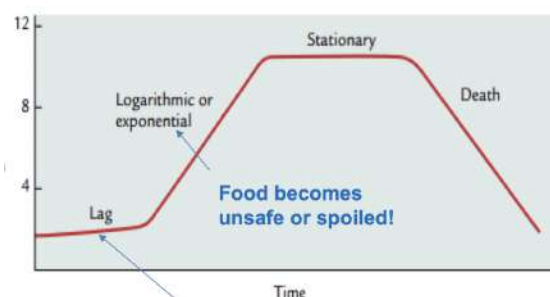


Figure 11: 4 Phase of Microbial Growth

16. Foodborne Illness

(a) Foodborne Illness = Food Poisoning

- Cause by contamination of food
- Can occurs at any stage of food production and consumption chain

(b) Ingestion of Contaminated Food and water

- Food Containing **living microbes** → **Foodborne Infection**
- Infection : Infectious Pathogens grows in the body
⇒ E.g. Bacteria, Viruses, Parasites
- Food Containing **living microbes** → **Foodborne Toxicoinfection**
- Toxicoinfection : Infectious Pathogens grows in the body
→ Produce toxin **in guts (Intestine)**
- Food containing **Pre-formed Toxin** → **Foodborne Intoxication** = Already have toxin in food before entering our body
⇒ Intoxication : Toxin causes illness
- Toxic Product of microbes : Chemical Substances
⇒ E.g. Mycotoxin

17. Foodborne Infection

(a) Cause by **Ingestion of Living Pathogens**

⇒ E.g. Bacteria, Prasites, Viruses

(b) Pathogens multiply in **Intestinal Mucosa**

(c) Examples

- Most Bacteria : *Salmonella spp.*, *Vibrio parahaemolyticus*, *Listeria monocytogenes*, *Campylobacter*
- All Parasites : *Giardia*, *Cyclospora*, *Cryptosporidium*
- All Viruses : *Norovirus*, *Hepatitis A*.

(d) Common Symptoms : **Diarrhea, Abdominal Cramps, Nausea, Fever**

18. Foodborne Toxicoinfection

(a) Cause by **Ingestion of Living Pathogens**

⇒ Can **produce toxins in the intestine**

(b) Toxins

- Damage intestinal cells
- Some travel through blood stream
⇒ Affect **Kidneys & Central Nervous System**

(c) Examples

- Shiga toxin-producing *E. coli* in intestine
⇒ E.g., *E. coli* O157H7
- *Shigella spp.*
- *Clostridium perfringens*
- *Bacillus cereus* (Diarrheal Type)

(d) Common Symptoms : **Diarrhea, Abdominal Cramps, Nausea, Vomiting, Fever**

19. Foodborne Intoxication

(a) Caused by ingestion of toxin in food

(b) Don't need to grow in gut

⇒ Fast Infection

(c) Bacteria grow before you ate

⇒ Produced toxin on the food

⇒ Poor Food Storage → Microbial Growth → Produce Enough Toxin

(d) Examples

- *Staphylococcus aureus* : Enterotoxins
⇒ Heat Stable
 - *Bacillus cereus* : Emetic Toxin
⇒ Heat Stable
 - *Clostridium botulinum* : Botulinum Toxin
⇒ Heat Labile = Sensitive to Heat
- (e) Heat Stable = Cannot be degraded under heating
⇒ Cannot remove toxins
- (f) Common Symptom : **Vomiting, Abdominal Cramps, Nausea, Diarrhea**
⇒ Seldom have **Fever**
20. Incubation Period : Foodborne Intoxication (Few Hours) < Foodborne Toxicoinfection < Foodborne Infection (Hours - Several Days)
∴ Toxin already present in food

2 Lecture 2 Detection and Enumeration of Microbes in Food I

1. Food Quality : Degree of Excellence of Maintaining Food
 - Safety
 - Shelf-life (Acceptability)
 - Consistency
2. We want to determine the types and numbers of microorganisms in food
3. Purpose of Microbiological Testing
 - (a) Meet the Microbiological Levels
 - Standards Criteria : Governmental
⇒ E.g. FDA's Bacteriological Analytical Manual (BAM), FSIS's Laboratory Guidebook
 - Standards Criteria : Scientific / Commodity Association
⇒ AOAC International
 - Standards Criteria : International
⇒ Codex Alimentarius, ISO
 - Guidelines : **Recommended** levels
⇒ Monitoring Microbiological Acceptability
 - **Specifications** : Commercial Contractual Condition of Acceptance
 - (b) Predict Product Shelf-life
⇒ $10^5 - 10^7$ = Start damaging the food (Spoilage is occurred) = Too late
 - (c) Assess Potential Safety
⇒ High microbial number → ↑ Risk of Pathogen + ↑ Risk of toxin present
 - (d) Monitor Hygiene Status
⇒ Need to **check microorganisms metabolize level**
∴ Microorganisms can produce toxin
4. Basic Steps in Microbiological Analysis

(a) Sampling

- Pick your Sample Solid Food / Liquid Food
- Equipment : Blender, Stomacher and Stomacher Bag

(b) Amplification

- Enrichment Cultures
- Immunomagnetic Separation (IMS)
- PCR
- Used to **Magnified the Sample**
∴ It might contain less sample

(c) Detection (Present / Absence)

- Selective Medium
⇒ E.g. Blood Agar
- Electrophoresis
- Enzyme Linked Immunosorbent Assays (ELISA)

(d) Confirmation (Identity & Activity)

- Serotyping : Agglutination Tests
⇒ Antigen on beacteria / virus can bind to their specific antibodies → **Aggregate + Form Clumps**



Figure 12: Serotyping : Agglutination Tests

- API Strip (Convenient Method)
- DNA Fingerprinting
⇒ Used to check the identity of bacteria

5. Sample Collection

- Sample = **Mass of Material being Examined**

- Method used should protect against microbial contamination
⇒ Microbial Numbers should not change between collection and analysis

6. Collecting Sample Procedural

- (a) Use **Individual Container** of food
- (b) Apply Aseptic technique
- (c) Sterile Sampling Instruments / Containers
- (d) Proper Labeling of all containers
- (e) Analyze quickly
- (f) Refrigeration and Freezing (Optional)

7. Traditional Culture Methods

- Medium Component : Nutrients, Salt, pH colour Indicator, Essential growth Factor, Specific Growth Inhibitor)
 - Amino-nitrogen : Peptone, Protein Infusion
 - **Growth Factor** : Blood, Serum, Vitamins
 - Energy Source : Sugar, Alcohols, Organic Acids
 - Buffer Salts : Phosphates, Acid Salts
⇒ Maintain pH
 - Mineral Salts & Metals : Phosphates, Sulphates, ...
⇒ Coenzyme & Cofactors
 - **Selective Agents** : Chemicals, Antibiotics, Bile Salts, Dyes, Antimicrobials
⇒ Specific to bacteria → We need to know the identity of bacteria in mixed bacteria before we use it
 - Indicator Dyes : Phenol Red, Neutral red
 - Gelling Agents : Agar, Gelatin, Gel
- Microorganism Growing Media
 - (a) Nutrient Broth (Liquid)
 - Used For Culturing & Enrichment
 - Bacteria Growth → Cloudy Solution (Turbidity)
⇒ Use **Spectrometer** to check Optical Density (OD)
⇒ This method is not accurate

(b) Nutrient Agar (Solid)

- Used for **Counting Numbers and Isolation of Single Colony**
- Observe Colony Morphology
- Agar = Broth + Gelling Agent → Solidified

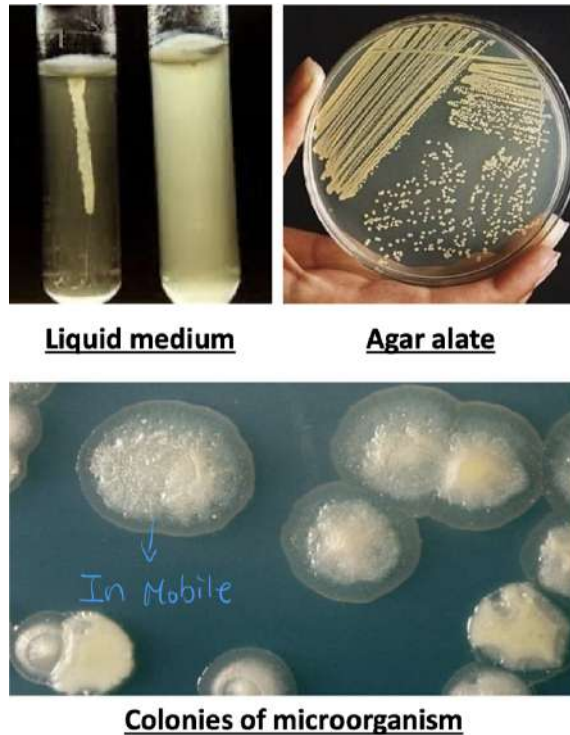


Figure 13: Growing Media

8. Culture Medium Hypothesis: At least one cell can form a colony or produce growth in the tube
⇒ Not always valid
∴ Cell Injury : Cannot grow and Die on the selective media under stress
∴ Viable but Non-culturable (VNC) State : Cells are still alive but cannot grow any more in new culture media
9. VNC State might underestimate the sample bacteria number
∴ Cannot grow in culture media doesn't mean that they cannot

grow in food

10. Prevention & Reach the Assumption : Make sure the incubation time is enough → Bacteria have enough time to adapt the new culture environment

11. Different Kinds of Culture Medium

- Plate Count Agar (The Most Common one) : Aerobic Mesophilic Count
- Crystal Violet / Blood Agar : Enumeration of **Fecal Streptococci**
⇒ Contain very High Nutrient → Used for Enrichment
- Thiosulfate / Citrate / Bile / Sucrose Agar (TCBS) : Isolation of **vibrios**
⇒ Used to **Differentiate** the bacteria type
⇒ E.g. *Vibrio cholerae* (Yellow) vs *Vibrio parahaemolyticus* (Green)
- MacConkey Agar (MAC) : *E. Coli* O157
⇒ E.g. **Non-lactose** Fermenting Colonies (Colourless) vs **Lactose** Fermenting Colonies (**Pink**)

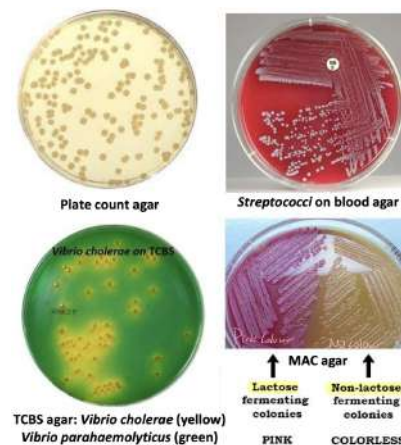


Figure 14: Type of Agar

12. Different Types and Purposes of Culture Medium

(a) Non-Selective

- For General Purpose
- No Selective Agents is needed to add
- **Monitor Total Bacterial Growth**
- Example 1 : **Plate Count Agar**
- Example 2 : **Luria-Bertani broth Agar (LB)**



Figure 15: Non-Selective Agar

(b) Enrichment

- Contain Extra Nutritional Value & Extra Growth Factors
⇒ Favor the Growth of Specific
⇒ Does not inhibit the growth of competing ones
- Example 1 : **Blood Agar (BAP)** for Microbes need blood to grow
- Example 2 : **Chocolate Blood Agar (CBA)**



Figure 16: Enrichment Agar

(c) Selective

- Allows Specific Microorganisms to Grow

- Suppress the growth of other microorganisms that we don't need
- But still cannot distinguish the type of bacteria
∴ It can **allow more than one microorganisms to grow in this agar**
⇒ We can only inhibit most of the microorganisms but not all
- Example 1 : ***Xylose lysine deoxycholate(XLD)* Agar** for ***Salmonella*** and ***Shigella*** Species
- Example 2 : **TCBS Agar** for isolating all ***Vibrio***



Figure 17: Selective Agar

(d) Differential

- It can distinguish different types of microorganisms **Based on Metabolic Differences**
⇒ It is **Specific**
- Example 1 : **Motile Agar** is used to distinguish microorganisms **Based on their Motility**
- Example 2 : **MacConkey (MAC) Agar** is used to differentiate the microbes **Base on Lactose Fermentation ability**

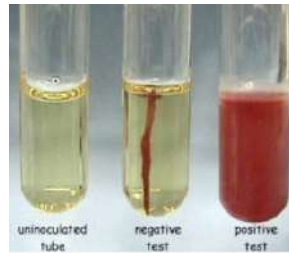


Figure 18: Differential Agar

13. Bacterial Colony

- Single Bacterial Cell is invisible on agar plate
- When single bacterium is fixed on agar → Can only grow in the specific location dissolved in the agar
- A single bacterium replicates and produces many daughter cells
 - ⇒ Can aggregate in specific locations to form a **Visible Mass Colony**
 - ⇒ 1 Colony = Many Bacterial Cells which are all from 1 Single Mother Cell
- 1 Colony = 1 Colony Forming Unit (CFU)
- This process takes time to grow

14. Different Bacteria has different properties

⇒ Can be identified by looking at the colony morphology

15. Colony Morphology of Bacteria

(a) *Staphylococcus aureus*

- Colony Shape : Circular Form
- Colony Margin : Entire Margin
- Optical Property : Opaque (Not Transparent)
- Colony Elevation : Convex
- Colony Colour : Golden Yellow
- Colony Surface : Smooth

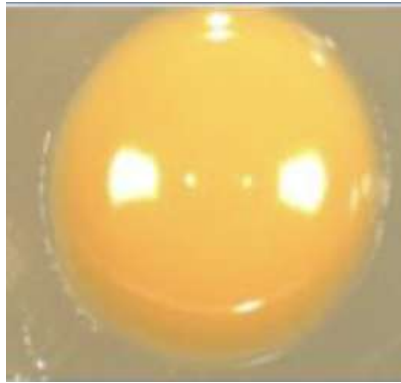


Figure 19: *S. aureus*

(b) *Streptococcus pneumoniae*

- Colony Shape : Circular Form
- Colony Margin : Entire Margin
- Optical Property : Opaque (Not Transparent)
- Colony Elevation : Raised
- Colony Colour : White
- Colony Surface : Smooth



Figure 20: *S. pneumoniae*

(c) *Pseudomonas aeruginosa*

- Colony Shape : Circular Form
- Colony Margin : Entire Margin
- Optical Property : Opaque (Not Transparent)
- Colony Elevation : Flat

- Colony Colour : Grey
- Colony Surface : Smooth



Figure 21: *Pseudomonas aeruginosa*

(d) *Mycobacterium tuberculosis*

- Colony Shape : Irregular Form
- Colony Margin : Undulate Margin (Wave Liked)
- Optical Property : Opaque (Not Transparent)
- Colony Elevation : Pulvinate (Pillow Liked)
- Colony Colour : Yellow
- Colony Surface : Rough



Figure 22: *Mycobacterium tuberculosis*

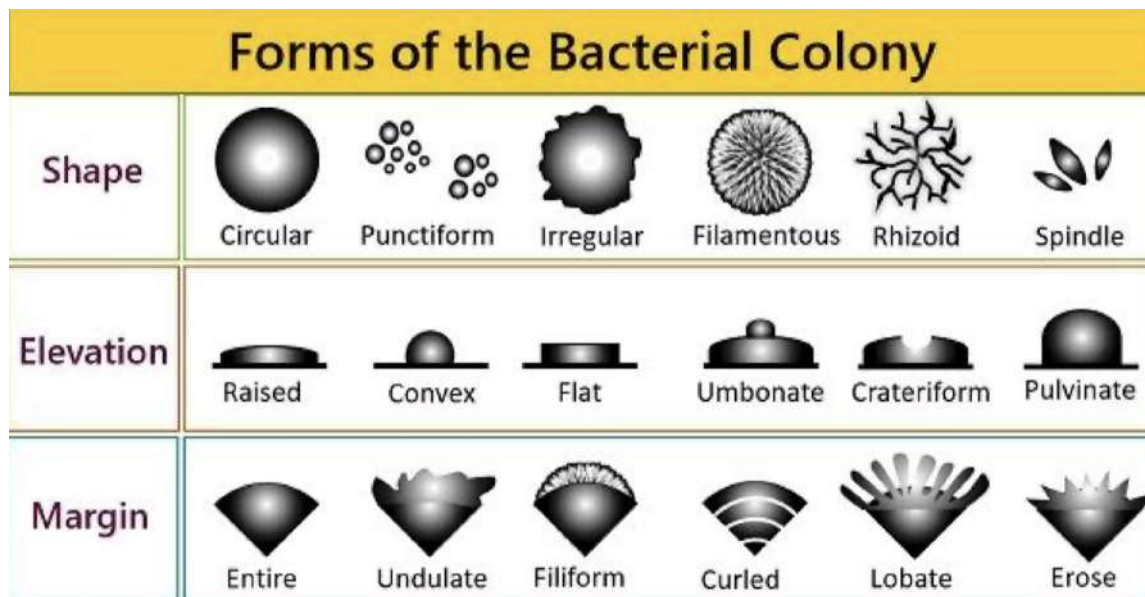


Figure 23: Forms of Bacterial Colony

3 Lecture 3 Detection and Enumeration of Microbes in Food II

1. Quantitative Culture Methods

- Plate Count
- Most Probable Number (MPN)
- Spiral Plate Count
- Membrane Filtration



Figure 24: Culturing Methods

2. Plate Count

- **Colony Forming Unit (CFU)** : Measure the number of **Viable Bacterial / fungal Cells** in a Sample / in terms of colonies
⇒ Liquid Sample Unit : CFU/ml
⇒ Solid Sample Unit : CFU/g
- **Standard Plate Count (SPC)** : Pour Plating
⇒ 1 ml pipetted directly into **Sterile Petri Dish** → Mixed with **Molten Agar** in temperature 40°C – 45°C
⇒ Thermal Shocked Psychrotrophs (Love Cold Bacteria) cannot make a visible colony in this method
- **Aerobic Plate Count (APC)** : Surface Plating
⇒ Sample Size : Maximum 0.1ml ⇒ Aerobic Environment is needed = Need to breathe in the incubator

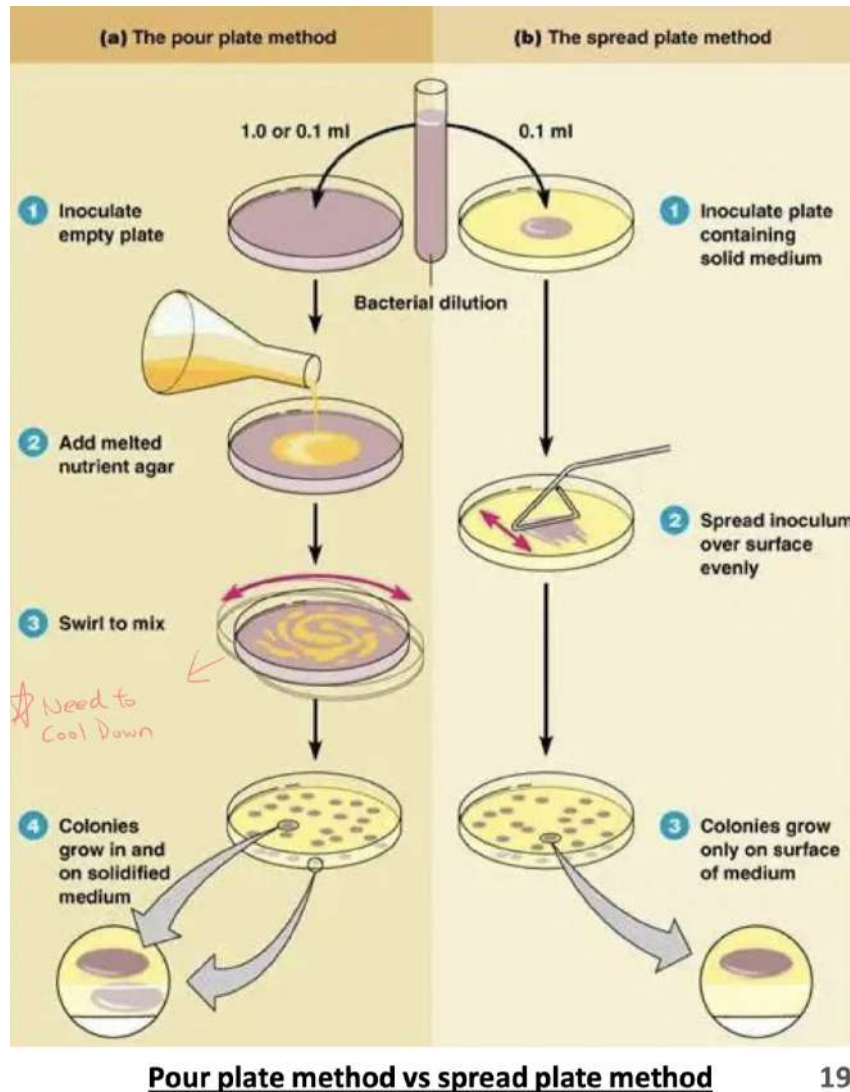


Figure 25: Step of Pour Plate & Spread Plate Method

3. Plate Counting Steps : **Homogenization** → **Serial Dilution** → **Plating** → **Incubation** → **Colony Counting**

4. Homogenization

- Mix the sample with **Diluent**
⇒ Bathing the Microorganism
- Produce a **Homogeneous Solution** by **Mechanical Blender / Stomacher**
⇒ Stomacher : Mixing the sample

- Calculation of Dilution Factor : $\frac{\text{Sample (g)}}{\text{Diluent Solution (ml)}}$

5. Serial Dilution

- Sample is diluted at **10 Fold Serially** → Then plating onto agar for **Subsequent Detection & Enumeration**
- 1st Dilutions from Original Inoculum : 1:10
 - ⇒ Transferring **1 ml Original Inoculum** into 1st Tube
 - ⇒ Originally, there exist **9 ml Broth** in 1st tube
 - ⇒ In Total, Tube 1 has 10 ml Mixed Dilutions
- 2nd Dilutions from 1st Dilution Tube : 1:100
 - ⇒ Transferring **1 ml 1st Tube Dilutions** into 2nd Tube
 - ⇒ Originally, there exist **9 ml Broth** in 2nd tube
 - ⇒ In Total, Tube 2 has 10 ml Mixed Dilutions
- Repeat this process 5 times
 - ⇒ Final Dilution Factor Ratio : 1:100000
- Then transfer 1 ml to the agar plate for culturing
- Use **Pour Plating Method (SPC)**

6. Plating

(a) Surface Plating

- Add **At Most 0.1 ml** of selected dilutions onto the Agar Surface
- Spread over with a glass spreader
- Do not pour 1 ml dilution onto one agar plate
∴ Solid Agar cannot absorb

(b) Pour Plating

- Add **At Most 0.1 ml** of selected dilutions onto the Surface of a Sterile Plate
- Add Sterile Molten (Warm) Agar (45°C)
- Mix the agar with the sample
- Allow the agar to solidify

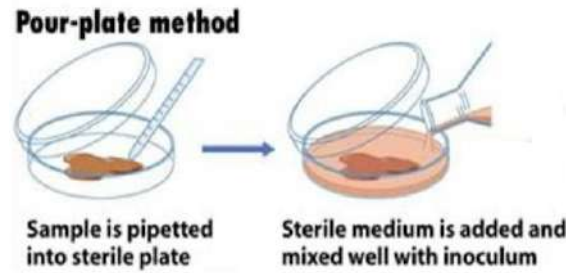


Figure 26: Pour-Plate Method

7. Incubation

- Incubator the plate **Upside Down in Incubator**
 $\Rightarrow$ Upside Down : Trap water droplet on the lid
- If not inverted upside down $\rightarrow$ Water droplet would fall to the agar surface $\rightarrow$ Disturb Microorganisms Growing during incubation
- Solid Agar Plate : Do not **sealed** the plate
- Liquid Medium : Cap Remain Loose State (Not in Lock State)
 $\therefore$ Microorganisms need to breath
 $\Rightarrow$ Only seal / Lock the plate / tube before fridge and storage
 $\Rightarrow$ Sealed Plate : Can avoid **Contamination** and **Plate Dry Out**

8. Colony Counting

- Count **Isolated Colonies**
 $\Rightarrow$ Only count 25 – 250 per plate = Statistically Significant
 $\Rightarrow$ Don't need to count that is out of this range
- 1 Count = 1 Cell / a clump of cells
- Unit of Number of colonies = CFU/ml or CFU/g
- Each dilution was applied in **Duplicate / Triplicate (2-3)** replicates and calculate the average count
 $\Rightarrow$ More Accurate Count

9. CFU Equation

$$N = \frac{\sum c}{(n_1 + 0.1 \times n_2) \times d}$$

- N = Number Count in cfu/g
- $\sum c$ = Sum of the Colonies Count on all Plates
- n_1 = Number of Plate in the **First Dilution**
- n_2 = Number of Plates in the **Second Dilution**
- d = Dilution Factor corresponding to the **First Dilution**
 $\Rightarrow$ If the **First Dilution** is not 1 ml / 1 g
 $\rightarrow d$ = Inoculated aliquot $\times$ Dilution Factor Corresponding to the **First Dilution**

10. Aerobic Plate Count (APC) Rounding Rule

- Change Calculated Counts (N) $\rightarrow$ Aerobic Plate Count (APC)
- **Only Look at the FIRST TWO SIGNIFICANT DIGITS**
- **Round Up the Second Digit : When Third Digit = 6, 7, 8, 9**
 $\Rightarrow$ Rounding Up = Right Position Digit are all becoming 0 + Second Digit + 1
- **Round Down the Second Digit : When Third Digit is 1, 2, 3, 4**
 $\Rightarrow$ Rounding Down = Right Position Digit + Second Digit are all becoming 0
- **Third Digit = 5 :**

{	Round Up	Second Digit = Odd
{	Round Down	Second Digit = Even
- After Rounding, APC need to **Round to 2 Significant Figures**
 $\Rightarrow$ Unit : cfu/g

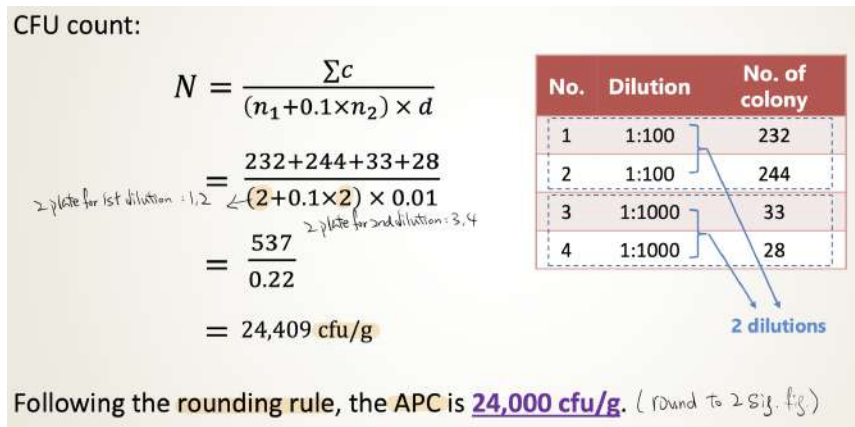


Figure 27: Example of APC Rounding

11. Plates with < 25 CFU

- Small Errors in dilution technique / Presence of small Contamination can make great effect on the final count
- If all plates from two dilutions are < 25 CFU each $\Rightarrow$ Then Record Number of Colony = **Too Few To Count (TFTC)**
- Record **Estimated Aerobic Plate Count (EAPC)** : $< 25 \times \frac{1}{d}$
 $\Rightarrow d$ = The Dilution Factor corresponding to the **First Dilution**

12. Plated with > 250 CFU

- Poor Isolation and Colonies will have grown together
- If plates from both 2 dilutions yield more than 250 CFU each $\Rightarrow$ Then Record Number of Colony = **Too Numerous To Count (TNTC)**
- Record **Estimated Aerobic Plate Count (EAPC)** : $> 250 \times \frac{1}{d}$
 $\Rightarrow d$ = The Dilution Factor corresponding to the **First Dilution**

13. Estimated Aerobic Plate Count (EAPC) : $c \times \frac{1}{d}$
 $\Rightarrow$ Unit : CFU/ml

4 Lecture 4 Detection and Enumeration of Microbes in Food III

1. Most Probable Number (MPN)

- Measure the number of viable bacterial cells in a **liquid sample**
- Base on statistical probability theory
⇒ Not the actual count observed on agar
- Used when sample contain **low concentration of bacterial cells**
⇒ < 30 CFU/ml
- Used when sample contain special material and can interfere the **standard plate count**
⇒ E.g. Milk, Potato Water
- We want to dilute the sample to the degree that contain viable organism
⇒ Degree : Able to grow and initiate activity in tube
⇒ Turn Turbid = +ve Result = Positive Tube
⇒ Clear = -ve Result = Negative Tube
- Result is less precise than the Standard Plate Count

2. Procedure of MPN

- (a) Commonly use ***E. coli* & coliforms**
- (b) Samples are serially diluted to the point of extinction
⇒ Point of Extinction = No Microorganisms = Clear Tube
- (c) The serial dilutions must contain at least 3 dilutions

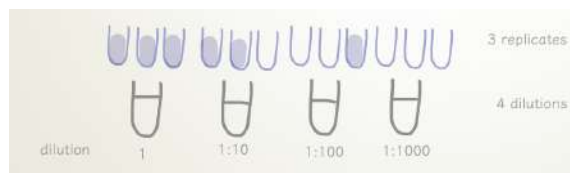


Figure 28: Dilution in MPN

- (d) Inoculate into **3 or 5** test tube of liquid media per dilution

- (e) Observe the tubes that show positive reaction
 $\Rightarrow$ Turbidity / Colour Change / Gas Formation
- (f) The most probable number of microorganisms is found by the **Standard MPN Table** and looking at the percentage of the tubes
 $\Rightarrow$ Percentage of the tubes = Numbers of tubes showing turbidity
- (g) Point of Extinction could not exist and it can still estimate the number of microorganisms

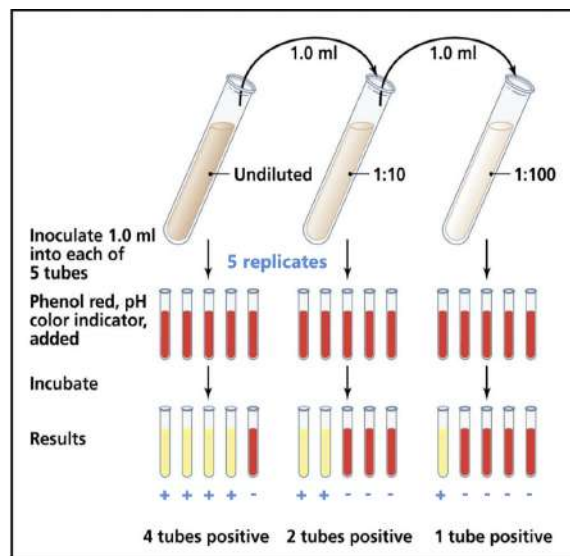


Figure 29: Inoculation in MPN

3. Most Probable Number Count

- (a) Three dilution are chosen to give good approximation to MPN
- (b) Rule 1 : More than 3 dilution $\rightarrow$ Choose 3 of them
- (c) Rule 2 : When the next dilution and itself have all negative tubes (Clear Tubes) $\rightarrow$ Remove the Highest Dilution

Examples	10 g	1 g	0.1g	0.01g	0.001g	3 dilutions
A	4	1	0	0	0	410xx

Figure 30: Rule 2

- i. 0.001 g is Negative Tube + 0.01 g is Negative Tube (Next Lower Dilution)
 - ii. Cancel 0.001 g tube
 - iii. 0.01 g is Negative Tube + 0.1 g is Negative Tube (Next Lower Dilution)
 - iv. Cancel 0.01 g tube
- (d) Rule 3 : When > 3 dilutions $\rightarrow$ Use the **Highest Dilution** with all positive tubes + Choose the **Three Highest Remaining Dilutions**

Examples	10 g	1 g	0.1g	0.01g	0.001g	3 dilutions
B	5	5	1	0	0	x510x
C	4	5	4	5	1	xx451

Figure 31: Rule 3

- Example B
 - i. 1 g is the most diluted with all positive tubes
 - ii. 0.001 g is canceled by **Rule 2**
 - iii. 1 g, 0.1 g, 0.01 g are the three highest remaining dilutions
 - Example C
 - i. 1 g and 0.01 g has the same positive value
 - ii. Pick 0.01 g
 $\therefore$ 0.01 g is the highest dilution
 - iii. 0.1 g, 0.01 g, 0.001 g are the three highest remaining dilution
- (e) Rule 4 : When > 3 dilutions + The highest dilution is **not within the three most diluted tubes** $\rightarrow$ Find the **Highest Dilution** with **all positive tubes** $\rightarrow$ Pick the **next two higher dilutions** + Third one Choose the **Higher Dilution Tube than the two chosen tubes**
 $\Rightarrow$ If more than 1 higher dilution tube $\rightarrow$ **The Third Tube Follows Rule 5**

NOT within the 3 highest remaining tube

Examples	10 g	1 g	0.1g	0.01g	0.001g	3 dilutions
D	4	5	4	3	1	xx431

Figure 32: Rule 4

- i. 1 g has the highest dilution with all positive tubes
 - ii. 1 g is not within the three most diluted tubes
 - iii. 0.1 g, 0.01 g are picked
 - iv. The third dilution is followed **A Higher Dilution from 0.01 g**
 $\Rightarrow$ 0.001 g will be chosen
- (f) Rule 5 : When > 3 dilutions + **No dilution with all positive tubes** $\rightarrow$ Choose the **Two Lowest Dilutions** + Sum up the positive tubes of all higher dilutions to the **third dilution**

Examples	10 g	1 g	0.1g	0.01g	0.001g	3 dilutions
E	4	3	0	+ 1	+ 1	432xx

Figure 33: Rule 5

- i. 10 g and 1 g are the two lowest dilutions
 - ii. All dilution are not all positive
 $\Rightarrow$ In this example, None of them get 5 positive tubes
 $\therefore$ We are doing 5 inoculating tubes in each dilution $\Rightarrow$
 Number of Positive Tube $\neq$ Number of Inoculating Tube
 - iii. **Sum up the positive tubes of the higher dilutions** $\rightarrow$ Become the third dilution
- (g) Rule 6 : When 3 dilutions selected are not in the table $\rightarrow$ Redoing the test / Use the **3 Highest Remaining Dilutions**

Examples	10 g	1 g	0.1g	0.01g	0.001g	3 dilutions
F	4	3	3	2	1	xx321

Figure 34: Rule 6

- i. All dilution are not all positive tubes
 $\Rightarrow$ None of them get 5 positive tubes

$\Rightarrow$ Number of Positive Tube $\neq$ Number of Inoculating Tube

- ii. By **Rule 5**, choose the two lowest dilutions (10 g and 1 g)
+ Sum up 0.1 g, 0.01 g, 0.001 g to be the third dilution
- iii. We can get positive tubes : 4-3-6 (Not in the table)
 $\Rightarrow$ **Rule 5 Fails** $\rightarrow$ Use Rule 6
- iv. Choose the highest dilution with any positive tubes
 $\Rightarrow$ Dilution : 0.1 g, 0.01 g, 0.001 g
 $\Rightarrow$ Positive Tubes : 3-2-1

(h) **No dilution has all positive tubes**

- Each number here are doing 3 / 5 inoculating tubes
 $\Rightarrow$ Number of Serial 10-fold Dilution does not have restriction
- In all of the example, we are doing 5 dilutions and 5 inoculating tubes
- If Number of inoculating tubes = Number of Positive Tube, we say it is all positive tubes
 $\Rightarrow$ In our example, only getting 5 positive can say all positive tubes
- If Number of inoculating tubes $\neq$ Number of Positive Tube, we say it is not all positive tubes

4. Standard MPN Table

- Three Dilutions are chosen
- Unit : MPN/mL or MPN/g = MPN causing positive tubes per unit volume of the original sample
- If three chosen tube dilution is not in 0.1, 0.01, 0.001 (In Figure 35), Then **Need to do Multiplication on the MPN/g value**
 $\Rightarrow$ If it is 0.01, 0.001, 0.0001 $\rightarrow$ MPN/g need to **multiply 10 (10-fold Difference)**
 $\Rightarrow$ Opposite (Division) if the dilution is lower

Answer **MPN table for 3 tubes**

Positive Tubes			MPN/ g	C.I.		Positive Tubes			MPN/ g	C.I.	
0.1	0.01	0.001		Low	High	0.1	0.01	0.001		Low	High
0	0	0	< >	-	9.5	2	2	0	21	4.5	42
0	0	1	3.0	0.15	9.6	2	2	1	28	8.7	94
0	1	0	3.0	0.15	11	2	2	2	35	8.7	94
0	1	1	6.1	1.2	18	2	3	0	29	8.7	94
0	2	0	6.2	1.2	18	2	3	1	36	8.7	94
0	3	0	9.4	3.6	38	3	0	0	23	4.6	94
1	0	0	3.6	0.17	18	3	0	1	38	8.7	110
1	0	1	7.2	1.3	18	3	0	2	64	17	180
1	0	2	11	3.6	38	3	1	0	43	9	180
1	1	0	7.4	1.3	20	3	1	1	75	17	200
1	1	1	11	3.6	38	3	1	2	120	37	420
1	2	0	11	3.6	42	3	1	3	160	40	420
1	2	1	15	4.5	42	3	2	0	93	18	420
1	3	0	16	4.5	42	3	2	1	150	37	420
2	0	0	9.2	1.4	38	3	2	2	210	40	430
2	0	1	14	3.6	42	3	2	3	290	90	1000
2	0	2	20	4.5	42	3	3	0	240	42	1000
2	1	0	15	3.7	42	3	3	1	460	90	2000
2	1	1	20	4.5	42	3	3	2	1100	180	4100
2	1	2	27	8.7	94	3	3	3	> 1100	420	-

Figure 35: Standard MPN Table for 3 Tubes

- Advantages
 - Simple to run and read
 - Can be selective
 - Good for **anaerobes**
 - Good for low concentration
- Disadvantage
 - Large volume of glassware and space
 - Lack of accuracy

MPN table for 5 tubes

Pos. Tubes			Conf. lim.			Pos. tubes			Conf. lim.		
0.1	0.01	0.001	MPN/g	Low	High	0.1	0.01	0.001	MPN/g	Low	High
0	0	0	<0	—	6.8	4	0	2	21	6.8	40
0	0	1	1.8	0.09	6.8	4	0	3	25	9.8	70
0	1	0	1.8	0.09	6.9	4	1	0	17	6	40
0	1	1	3.6	0.7	10	4	1	1	21	6.8	42
0	2	0	3.7	0.7	10	4	1	2	26	9.8	70
0	2	1	5.5	1.8	15	4	1	3	31	10	70
0	3	0	5.6	1.8	15	4	2	0	22	6.8	50
1	0	0	2	0.1	10	4	2	1	26	9.8	70
1	0	1	4	0.7	10	4	2	2	32	10	70
1	0	2	6	1.8	15	4	2	3	38	14	100
1	1	0	4	0.7	12	4	3	0	27	9.9	70
1	1	1	6.1	1.8	15	4	3	1	33	10	70
1	1	2	8.1	3.4	22	4	3	2	39	14	100
1	2	0	6.1	1.8	15	4	4	0	34	14	100
1	2	1	8.2	3.4	22	4	4	1	40	14	100
1	3	0	8.3	3.4	22	4	4	2	47	15	120
1	3	1	10	3.5	22	4	5	0	41	14	100
1	4	0	11	3.5	22	4	5	1	48	15	120
2	0	0	4.5	0.79	15	5	0	0	23	6.8	70
2	0	1	6.8	1.8	15	5	0	1	31	10	70
2	0	2	9.1	3.4	22	5	0	2	43	14	100
2	1	0	6.8	1.8	17	5	0	3	58	22	150
2	1	1	9.2	3.4	22	5	1	0	33	10	100
2	1	2	12	3.4	22	5	1	1	46	14	120
2	2	0	9.3	3.4	22	5	1	2	63	22	150
2	2	1	12	4.1	26	5	1	3	84	34	220
2	2	2	14	5.9	36	5	2	0	49	15	150
2	3	0	12	4.1	26	5	2	1	70	22	170
2	3	1	14	5.9	36	5	2	2	94	34	230
2	4	0	15	5.9	36	5	2	3	120	36	250
3	0	0	7.8	2.1	22	5	2	4	150	58	400
3	0	1	11	3.5	23	5	3	0	79	22	220
3	0	2	13	5.6	35	5	3	1	110	34	250
3	1	0	11	3.5	26	5	3	2	140	52	400
3	1	1	14	5.6	36	5	3	3	180	70	400
3	1	2	17	6	36	5	3	4	210	70	400
3	2	0	14	5.7	36	5	4	0	130	36	400
3	2	1	17	6.8	40	5	4	1	170	58	400
3	2	2	20	6.8	40	5	4	2	220	70	440
3	3	0	17	6.8	40	5	4	3	280	100	710
3	3	1	21	6.8	40	5	4	4	350	100	710
3	3	2	24	9.8	70	5	4	5	430	150	1,100
3	4	0	21	6.8	40	5	5	0	240	70	710
3	4	1	24	9.8	70	5	5	1	350	100	1,100
3	5	0	25	9.8	70	5	5	2	540	150	1,700
4	0	0	13	4.1	35	5	5	3	920	220	2,600
4	0	1	17	5.9	36	5	5	4	1,500	400	4,600
5	5	5	>1600	700	—	—	—	—	—	—	—

Figure 36: Standard MPN Table for 5 Tubes

5. Coliform Bacteria

- A group of bacterial which **inhibit the intestinal tract** of animal in large number
- If they appears in water / food → Fecal Contamination
- Presence of Enteric (Intestinal) Bacteria : **Ferment Lactose**
⇒ E.g. *E. coli*, *Citrobacter*
⇒ All of them are similar → Not easy to distinguished
- Coliform = Gram-negative + Facultative Anaerobic + Rod Shape Bacteria (Bacilli) → Ferments Lactose → Produce Acid and Gas
⇒ Can do it within 48 hours & 35°C

6. Detection and Enumeration of Coliform Bacteria

- ≥ 3 different 10-fold serial dilutions
- Sample Dilutions $\xrightarrow{\text{Inoculation}}$ 3 / 5 Testing Tubes (Different Selective Medium) → Calculating **MPN**

⇒ Mark down the positive and negative result

- 3 Sets of Tests with Different Selective Medium : All are used to Detect Coliform

(a) Presumptive Test (LST)

- Confirm the Coliform Bacteria exist
∴ Coliform bacteria can ferment lactose
⇒ Produce acid and gas (In Durham Tube)
- Limitation : Some bacteria are gram positive

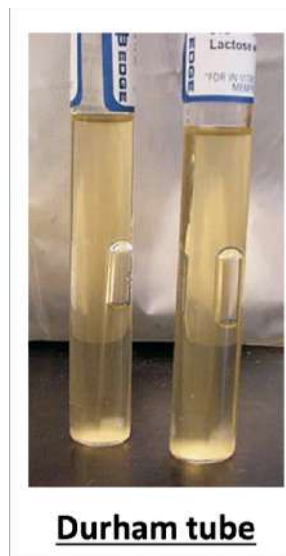


Figure 37: Durham Tube

(b) Confirmed Test (BGBB & EC)

- Use Broth Containing Inhibitor
- Can inhibit gram positive bacteria
⇒ Confirm the coliform bacteria exist

(c) Completed Test (Indole Test)

- Identify whether it is *E. coli* coliform

7. Different Test vs Coliforms Type

- (a) LST Test : MPN for Testing All Coliforms = MPN for Testing Presumptive Coliforms



Figure 38: Lauryl Sulphate Tryptose Broth

(b) BGGB Test : MPN for Testing Total Coliforms

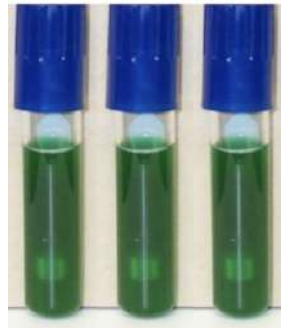


Figure 39: Brilliant Green Bile Broth

(c) EC Test : MPN for Testing Fecal Coliforms



Figure 40: *Escherichia coli* Broth

(d) Indole Test : Testing *E. coli*

- Positive Result : A Red Layer on the Borth Surface = Positive Result
- Negative Result : Clear Broth Tubes = Negative Result

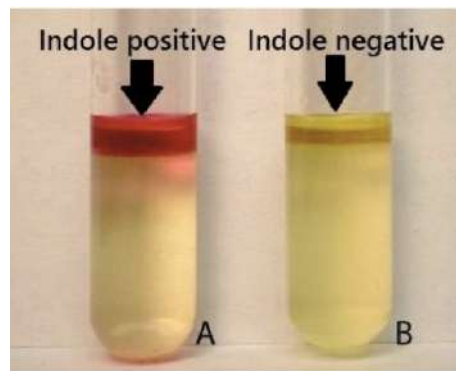


Figure 41: Indole Test

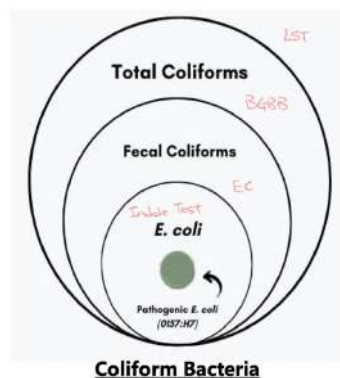


Figure 42: Coliform Bacteria Map

8. Steps of Coliform Bacteria Detection

- Preparing the Food Sample
- Homogenized = Blending the Food Sample
- Serial Dilution into 3 / 5 Test Tubes
- LST Tubes** : MPN for **Presumptive Coliforms**
- 3 Sub Pathways from LST Testing Tubes
 - BGBB Tubes : MPN for **Total Coliforms**
 - EC Tubes : MPN for **Fecal Coliforms**
 - Indole Test : Indole Positive vs Indole Negative
 $\Rightarrow$ ***E. coli***